

# HAOTIAN ZHENG

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Software engineer with expertise in SwiftUI and real-time graphics. Currently at Apple, building UI frameworks used across macOS and iOS. Previously created top-ranked indie apps. Passionate about human-computer interaction and creative tools.

## SKILLS

**Framework Dev** (SwiftUI, AppKit), **Application Dev** (iOS/macOS/Android, 2000+ stars on [GitHub](#)), **Game Dev** (Unity3D, Playdate), **Web Dev** (Vue.js), **Languages** (Swift/Objective-C/Java/C#), **UI Design** (Sketch), **Photography** (43M+ Unsplash views, 32 front-page features)

## EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA  
*Master of Science in Mobile and IoT Engineering (GPA 3.91/4.0)* *January 2021 - May 2022*
- **Central South University** Changsha, China  
*Bachelor of Engineering in Computer Science and Technology* *Sept 2014 - June 2019*

## PROFESSIONAL EXPERIENCE

- **Apple** Cupertino, CA  
*Senior UI Framework Engineer* *July 2022 - Present*
  - **Development:** Designed and implemented a dozen public SwiftUI APIs (Concentricity, BackgroundExtension, Window Styling, Outline Table, etc), working cross-functionally with Human Interface, technical writers, and adopter teams to ensure clarity and platform consistency. Named co-inventor on US patent applications.
  - **Ownership:** Led Interactive Liquid Glass on Mac across SwiftUI and AppKit. Served as the component owner of SwiftUI List, Table, and ScrollView on Mac. Responsible for issue triage, bug fixes, performance optimizations (up to 6x faster in cold launch time), test reliability, and roadmap planning for future improvements.
  - **Presentation:** Presented as the session speaker of Compose advanced graphics effects with SwiftUI in WWDC 2026 and Tailor macOS windows with SwiftUI in WWDC 2024, as well as in-person demos during WWDC at Apple Developer Center.
- **Apple** Cupertino, CA  
*Bluetooth Software Intern* *May 2021 - August 2021*
  - **Development:** Built internal Bluetooth audio prototype using CoreAudio, CoreBluetooth, SwiftUI, and XPC.
  - **Presentation:** Among top 10% of SWE interns to present project demos to Craig Federighi, SVP of SWE.
- **rct studio** Beijing, China  
*Co-founder & Chief Engineer* *September 2018 - January 2019*
  - **Development:** Led development of interactive VR pipeline, a REPL-like tool turning movie scripts into 3D visualizations.
  - **Collaboration:** Coordinated closely with other founders for the startup to be admitted to Y Combinator W19 batch.

## AWARDS & RECOGNITION

- **Apple WWDC Scholarships**
  - Won the 2022 one with Pegboard, a game logic node editor similar to UnrealEngine Blueprint but implemented purely in Swift. Designed the system to execute data flow and logic flow with an extensible data and UI model. (SwiftUI)
  - Won the 2020 one with Shader Node, an interactive GLSL shader playground. Implemented graph structure and searching algorithms. Introduced code generation for GLSL operators and types so shader previews can be updated in real-time. (SpriteKit, UIKit Dynamics, Combine, SwiftUI).
  - Won the 2018 one with Golf GO, a golf game under 1000 lines capable of generating millions of procedurally generated terrain maps. Achieved runtime mesh modification by applying noise values as vertex offsets. (SceneKit, GameplayKit, ModelIO).
- **Indie Sales & Media Coverage**
  - Developed, launched, and promoted Skyline on Android, once ranked as the top paid app on Google Play (US region, Jan 21, 2018) and covered by The Verge, Lifehacker, and The Next Web.
  - Published iOS apps that were featured on App Store (Apps We Love Right Now) and #4 Product of the Day on Product Hunt.
- **Hackathons**
  - Multiple hackathon winner record, including 1st place in IKEA hackathon, 1st place in SegmentFault Hackathon, 2nd place in Microsoft & HNU Hackathon, 2nd place in China Academy of Art Hackathon, 3rd place in Uber Hackathon, etc.

## SIDE PROJECTS

- **Unity Native Integration** *June 2017 - January 2021*
  - Published native plugins on Asset Store that enable Unity engine to dynamically display to multiple surfaces on Android. Utilized by multiple Unity developers & vendors across 100k+ users.
  - Deployed ContentProvider-based early-init techniques on Android and method swizzling on iOS to safely redirect and dispatch conditional lifecycle calls, greatly extending the engine's behavior for non-game use-cases.
- **Indie Applications** *October 2016 - January 2021*
  - Recreated Apple Music's fluid-like album effect in Android by implementing Kawase Blur and Domain Warping in LibGDX. Improved shader performance by 80% with a lookup table, as measured by RenderDoc / GAPID. Used by 200k+ users under the name of Diffuse.